

GRADIENT BOOSTING DECISION TREES FOR CREDIT SPREAD PREDICTION

Luka Edzgeradze, Caterina Federici, Emanuele Imperiali, Shervin
Nayebzadeh, Leonid Zhuravlev

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1 Introduction

Credit spreads play a central role in financial markets because they capture how investors perceive credit risk and overall economic conditions. Defined as the difference between corporate bond yields and risk-free Treasury yields, credit spreads tend to widen during periods of economic stress and narrow when financial conditions improve. As a result, movements in credit spreads are closely linked to business cycles, market volatility, and changes in monetary policy.

Because credit spreads aggregate information about risk aversion, corporate balance sheet health, and financial conditions, their dynamics are of direct interest to investors, portfolio managers, and policymakers. Predicting changes in credit spreads can help with risk management, help with decisions about where to put money, and help with decisions about how to finance a business. For policymakers and regulators, early signs of widening spreads could mean that financial conditions are getting worse or that credit markets are starting to feel stress. This could mean that they need to keep a closer eye on things or take action.

The objective of this project is to develop a predictive framework for modeling and forecasting the spread between U.S. Treasury yields and CCC-rated corporate bond yields. The focus on CCC credit captures the behavior of lower-rated, higher-risk corporate debt, which is particularly sensitive to changes in economic conditions and market sentiment. The proposed framework integrates a broad set of macroeconomic indicators, market-based variables, and financial conditions data in order to provide a forward-looking view of credit risk dynamics.

To accommodate the nonlinear and state-dependent nature of credit spread movements, the project employs modern machine learning techniques based on gradient boosted decision trees. These models are well suited to capturing complex interactions between macroeconomic fundamentals, financial markets, and credit conditions. By combining rich economic information with flexible model architectures, the analysis aims to improve predictive accuracy and provide economically interpretable insights into the drivers of credit spread fluctuations.

2 Market Indicators

Role of Credit Spreads. Credit spreads indicate the cost that investors are required to incur for the risks associated with the possibilities of default and uncertainties concerning the overall economic conditions. The credit spreads get significantly impacted by the overall dynamics and monetary conditions. While working on this project, we have used the overall macro-economic indicators.

Indicator Grouping. The selected indicators are grouped into four main categories: market sentiment, economic conditions, monetary policy and interest rates, and credit conditions. Together, these variables allow the models to capture changes in risk aversion, business cycle fluctuations, and financial stress that drive movements in credit spreads.

Market Sentiment. Firstly, market sentiment indicators capture investors' perception of risk and uncertainty in financial markets. We also incorporate the VIX index, representing the expected equity market volatility, and the S&P 500 index, proxying for overall market performance and confidence. Times of high volatility and falling equity markets are normally characterized by greater risk aversion; hence, investors demand wider credit spreads.

Macroeconomic Fundamentals. Macroeconomic fundamentals are at the core of determining corporate default risk. We capture real economic activity and conditions in the labor market with the unemployment rate, GDP growth, industrial production growth, and inflation (CPI). Generally, weak economic conditions—like those associated with rising unemployment or slowing growth—relate to deteriorating corporate balance sheets and higher expected probabilities of default, and, therefore, to wider credit spreads. Conversely, strong growth and production tend to support tighter spreads.

Monetary Policy and Interest Rates. Furthermore, it has been found that the credit spreads are affected by the monetary policies because of the influence on the financing conditions and discount rates. The variables added include the Federal Funds Rate, the yields for the 2-year and 10-year U.S. Treasury, and also the yield slope based on the difference between the 10-year and 2-year yields. The yield slope feature has been recognized as a warning sign for a potential recession, indicating market perceptions of economic futures.

Credit Conditions. To specifically capture stress episodes in credit markets, variables related to corporate leverage, liquidity, and credit risk are included. These variables are corporate leverage measures, the growth of commercial paper, and the CCC credit spread. Variables that are related to poor credit fundamentals and associated with greater uncertainty, such as higher leverage ratios and adverse trends for sub-investment grade credit, are generally associated with increased credit spreads.

Overall, the model expects variables associated with higher levels of uncertainty and weaker economic fundamentals, such as VIX, unemployment, inflation, and debt, to have a positive impact on credit spreads, while variables reflecting stronger economic conditions and market confidence, such as GDP, industrial production, and stocks, to have a negative impact on credit spreads. The model is designed to handle the nonlinear characteristics by including macroeconomic variables.

3 Data Collection and Construction

This section describes the data sources, acquisition procedures, and preprocessing steps used to construct the macro-financial dataset employed in the credit spread prediction models. All data collection and initial cleaning steps are implemented programmatically to ensure transparency and reproducibility.

3.1 Data Sources

The dataset combines information from two primary public data providers:

- **Federal Reserve Economic Data (FRED):** Used for macroeconomic indicators, interest rates, and credit market variables.
- **Yahoo Finance:** Used for market-based financial variables, including equity indices and volatility measures.

All series are collected starting from January 1960, subject to individual series availability, and extend to the most recent observation at the time of data retrieval.

3.2 Credit Spread Construction

The primary target variable is the CCC–10Y credit spread, constructed as the difference between:

1. Moody’s Seasoned CCC Corporate Bond Yield (FRED series `BAMLH0A3HYCEY`), and
2. The 10-Year U.S. Treasury yield (FRED series `GS10`).

Both yields are originally reported in percentage terms and are converted to decimal form. The CCC yield is available at a daily frequency, while the Treasury yield is reported monthly. To ensure temporal alignment, the Treasury series is interpolated to a daily frequency using time-based interpolation. The resulting spread is defined as

$$\text{CCC–10Y Spread}_t = y_t^{\text{CCC}} - y_t^{10Y}. \quad (1)$$

3.3 Market Variables

Market-based indicators are sourced from Yahoo Finance. These include:

- The CBOE Volatility Index (VIX), using daily closing values.
- The S&P 500 index level, from which percentage returns are computed.

Daily price series are reindexed to a common daily calendar and interpolated to address non-trading days. All timestamps are converted to timezone-naïve formats to maintain consistency across sources.

3.4 Macroeconomic and Yield Curve Variables

A range of macroeconomic indicators is obtained from FRED:

- Short-term interest rates (2-Year Treasury yield, `GS2`), used in combination with the 10-Year yield to compute the yield curve slope.
- Labor market conditions via the unemployment rate (`UNRATE`).
- Inflation, proxied by the month-over-month percentage change in the Consumer Price Index (`CPIAUCSL`).
- Economic activity measures, including GDP growth (`GDP`) and industrial production growth (`INDPRO`).

Quarterly and monthly series are transformed into daily frequency by forward-filling or interpolating, after which growth rates are computed using first differences in logarithmic or percentage terms where appropriate. The yield curve slope is defined as

$$\text{YieldCurveSlope}_t = y_t^{10Y} - y_t^{2Y}. \quad (2)$$

3.5 Firm-Level and Credit Market Indicators

To capture firm-level and aggregate credit conditions, additional FRED series are incorporated:

- Commercial paper outstanding (`CP`), used to compute growth rates as a proxy for short-term corporate financing conditions.
- Total credit market debt outstanding (`TCMDO`), from which a corporate leverage growth measure is derived.

As with other lower-frequency series, these variables are reindexed to a daily frequency and interpolated to ensure alignment with the target variable.

3.6 Final Dataset Assembly

All transformed series are combined into one daily panel dataset that is organised by date. Time-based interpolation is used to fill in missing values that come from frequency conversion or differencing. The dataset that comes out of this will be used as input for the next steps in feature engineering, model estimation, and evaluation, which are explained in later sections. This systematic, source-transparent data collection process makes sure that macroeconomic, market, and credit variables are all consistent with each other. It also makes it possible to do strong empirical analysis of credit spread dynamics.

3.7 Summary of Variables and Data Sources

Table 2 and Table 1 provide a consolidated overview of all variables used in the empirical analysis, together with their definitions, original data sources, and native reporting frequencies prior to transformation.

Table 1: Credit, Market, and Yield Curve Variables

Variable Name	Description	Source	Frequency
CCC_10Y_Spread	CCC corporate bond yield minus 10Y Treasury yield	FRED	Daily / Monthly
CCC_Yield	Moody’s Seasoned CCC Corporate Bond Yield	FRED	Daily
T10Y	10-Year U.S. Treasury yield	FRED	Monthly
T2Y	2-Year U.S. Treasury yield	FRED	Daily
YieldCurveSlope	Term spread (10Y minus 2Y Treasury)	FRED	Daily
VIX	CBOE Volatility Index (close)	Yahoo Finance	Daily
SP500_Return	S&P 500 percentage return	Yahoo Finance	Daily

Table 2: Macroeconomic and Credit Condition Variables

Variable Name	Description	Source	Frequency
Unemployment	U.S. unemployment rate	FRED	Monthly
Inflation	Month-over-month CPI inflation	FRED	Monthly
GDP_Growth	GDP growth rate	FRED	Quarterly
IndPro_Growth	Industrial production growth rate	FRED	Monthly
CP_Growth	Commercial paper outstanding growth	FRED	Quarterly
CorporateLeverage	Growth in total credit market debt outstanding	FRED	Quarterly

All series are transformed to a common daily frequency through interpolation or forward-filling to facilitate joint modeling with the credit spread target variable. We found that using an interpolation on the less frequent data produced significantly better results. Although it can be considered erroneous from the economic point of view, this allowed the model to better grasp the trend movements for the future prediction. The growth rates and first differences are computed after frequency alignment to ensure temporal consistency between variables.

4 Gradient Boosting Methods

4.1 General Principle of Gradient Boosting

Gradient Boosting is an ensemble learning technique that builds a strong predictive model by sequentially combining multiple weak learners, in the case of our implementation de-

cision trees. Our main idea is to optimize a predictive function by minimizing a specified loss function using gradient-based methods in function space.

Let $\{(x_i, y_i)\}_{i=1}^N$ denote the training dataset, where $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$. Gradient boosting constructs an additive model of the form

$$\hat{y}_i = F_M(x_i) = \sum_{m=1}^M f_m(x_i), \quad (3)$$

where each f_m is a weak learner and M is the total number of boosting iterations.

At iteration m , the model is updated by adding a new learner that minimizes the loss function $L(y, F(x))$ in a greedy manner, where the next weak learner is selected by optimizing the current learner without touching previously fitted learners. This serves to avoid computationally inefficient high dimensional global optimization. Specifically, the algorithm computes the negative gradients, as such:

$$r_i^{(m)} = - \left. \frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \right|_{F=F_{m-1}}, \quad (4)$$

and fits a weak learner $f_m(x)$ to these residuals. The model is then updated as

$$F_m(x) = F_{m-1}(x) + \nu \cdot f_m(x), \quad (5)$$

where $\nu \in (0, 1]$ is the learning rate controlling the contribution of each learner.

This iterative procedure continues until convergence or until a predefined number of estimators is reached. In our optimisation, taking into account, by nature small available data set of quarterly or monthly reported indexes and macro-indicators, we chose not to limit the maximum depth of the estimators, but limited number of estimators to conservative 3000.

4.2 XGBoost Regressor

XGBoost (Extreme Gradient Boosting) is an optimized and regularized implementation of gradient boosting designed for high performance and scalability. In the regression setting, the model predicts continuous target values using an ensemble of decision trees.

Unlike classical gradient boosting, XGBoost explicitly defines an objective function that includes both a loss term and a regularization term:

$$\mathcal{L}^{(m)} = \sum_{i=1}^N L(y_i, \hat{y}_i^{(m)}) + \sum_{k=1}^m \Omega(f_k), \quad (6)$$

where $\Omega(f)$ is a regularization function that penalizes model complexity, typically defined as

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2, \quad (7)$$

with T being the number of leaves in the tree and w_j the leaf weights.

XGBoost uses a second-order Taylor expansion of the loss function to improve optimization efficiency:

$$L(y_i, \hat{y}_i + f(x_i)) \approx L(y_i, \hat{y}_i) + g_i f(x_i) + \frac{1}{2} h_i f(x_i)^2, \quad (8)$$

where g_i and h_i are the first and second derivatives of the loss with respect to $\hat{y}_i$. This second-order information allows XGBoost to perform more accurate tree splits and faster convergence. Additional features of XGBoost include shrinkage, column subsampling, row subsampling, and efficient handling of missing values, all of which contribute to improved generalization and computational efficiency.

4.3 LightGBM

LightGBM is a gradient boosting framework specifically designed to handle large-scale datasets with high efficiency. It follows the same general boosting principle but introduces key algorithmic innovations to reduce training time and memory usage.

One of the main distinctions of LightGBM is its use of a leaf-wise (best-first) tree growth strategy instead of the traditional level-wise approach. At each iteration, LightGBM splits the leaf that yields the maximum reduction in the loss function, which can be expressed as

$$\Delta \mathcal{L} = \frac{1}{2} \left(\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right) - \gamma, \quad (9)$$

where G_L, G_R and H_L, H_R are the sums of first- and second-order gradients for the left and right child nodes, respectively.

LightGBM uses histogram-based binning to make scalability even better. This turns continuous features into a set number of bins. This cuts down on memory use a lot and makes it easier to find splits. LightGBM also supports Gradient-based One-Side Sampling (GOSS), which keeps data points with big gradients and randomly samples those with small gradients. It also supports Exclusive Feature Bundling (EFB), which combines sparse features that don't overlap.

5 Model Specification and Implementation

This section documents the concrete machine learning models implemented in this study. The modeling framework focuses on predicting the level of the CCC-10Y credit spread using macroeconomic and financial variables, augmented with engineered temporal features. Two gradient boosting-based regression models are employed: *XGBoost* and *LightGBM*.

5.1 Feature Engineering and Data Preparation

The target variable is the level of the CCC-10Y credit spread. The input feature set consists of a base group of macro-financial indicators, including equity market volatility (VIX), equity index levels (S&P 500), macroeconomic growth indicators, inflation, unemployment, corporate leverage, and yield curve slope. To capture medium- and long-term dynamics, first differences of each base feature are computed over multiple horizons:

$$\Delta_k x_t = x_t - x_{t-k}, \quad k \in \{30, 60, 180, 365\}, \quad (10)$$

corresponding to monthly, bi-monthly, semiannual, and annual changes. In addition, rolling volatility measures of the credit spread are calculated over analogous windows using rolling standard deviations of first differences. The dataset is split chronologically into training, validation, and test sets to preserve the time-series structure. For the XGBoost model, all numeric features are standardized using a z-score transformation, while LightGBM operates directly on the raw numerical features without explicit scaling.

5.2 XGBoost Regression Model

The XGBoost regressor is configured as a high-capacity gradient boosting model with a large number of estimators and a small learning rate, reflecting a preference for incremental functional updates. Formally, the model takes the additive form

$$\hat{y}_i = \sum_{m=1}^M f_m(x_i), \quad (11)$$

where each f_m is a regression tree and $M = 3000$ boosting rounds. The model is trained using the squared error loss function,

$$L(y, \hat{y}) = (y - \hat{y})^2, \quad (12)$$

and optimized via second-order gradient boosting. Tree depth is unconstrained, allowing the algorithm to learn complex nonlinear interactions among engineered macroeconomic features. Early stopping is implicitly controlled through validation monitoring rather than explicit regularization constraints. This configuration yields near-perfect in-sample fit and very high out-of-sample explanatory power on the validation set. Considering the nature of the underlying data, with small increments of change from t_i to t_{i+1} , we prioritized letting the model discover close and complex relationships with the data, thus setting the depth of the estimators to unlimited.

5.3 LightGBM Regression Model

The LightGBM regressor is trained using the same target variable and feature set but leverages LightGBM’s histogram-based and leaf-wise tree construction strategy. The model objective is standard least-squares regression, and predictions take the form

$$\hat{y}_i = \sum_{m=1}^M g_m(x_i), \quad (13)$$

where each g_m corresponds to a leaf-wise optimized decision tree.

Key hyper-parameters include an unfixed number of leaves, a small learning rate (0.02). Tree depth is left unconstrained, with model complexity essentially unregulated. Though, obviously that endangers our model to become overfit, we prioritized hidden relationship detection by limiting the tree size. Through experimentation, this approach showed the best result, which was expected. Though unconventional, due to the fact that this model is to be specialized for this particular bond spread and not a general solution, we prioritise the accuracy and overlook the overfitting issue.

Unlike XGBoost, LightGBM directly consumes unscaled numerical inputs and internally discretizes features into histograms, enabling efficient split finding even with a large number of engineered features. This setup produces a slightly lower in-sample error and higher directional accuracy on validation data, at the cost of a modest reduction in R^2 relative to XGBoost.

5.4 Evaluation Metrics

Model performance is assessed using both level-based accuracy and directional consistency. Standard regression metrics include Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), coefficient of determination (R^2), and Mean Absolute Percentage Error (MAPE). Additionally, directional accuracy is computed by comparing the sign of successive changes in predicted and realized spread levels:

$$DA = \frac{1}{T-1} \sum_{t=2}^T \mathbb{I}[\text{sign}(y_t - y_{t-1}) = \text{sign}(\hat{y}_t - \hat{y}_{t-1})]. \quad (14)$$

This combination of metrics allows evaluation of both static level prediction accuracy and dynamic predictive usefulness for directional risk assessment, as illustrated by the validation results and diagnostic plots reported in the appendix. Furthermore, understanding the nature of the time-series financial data prediction on such small increments, we highly prioritised directional accuracy when making the evaluative conclusion, as on small scale, very small magnitude of prediction produces a seemingly accurate result, where in fact the opposite is true. The model just recognizes the magnitude of change from previous day and reassigns it, which holds no predictive value.

6 Practical Applications

Portfolio Management and Risk Monitoring. The results of this project have several practical applications for investors and risk managers. By providing forward-looking estimates of changes in corporate credit spreads, the model can serve as an early warning signal of rising market stress and deteriorating credit conditions. Investors can use these signals to adjust portfolio allocations between corporate bonds and safer assets, or to decide when to hedge credit exposure using instruments such as credit indices or bond ETFs.

Asset Allocation Decisions. For example, consider an investor managing a portfolio allocated across investment-grade corporate bonds, high-yield bonds, and U.S. Treasuries. If the model forecasts widening credit spreads, this signals rising credit risk and deteriorating market conditions. In response, the investor might reduce exposure to corporate bonds, particularly lower-rated credit, and reallocate toward Treasuries or cash to protect the portfolio from potential losses. Alternatively, the investor could maintain credit exposure while hedging risk using credit indices or bond ETFs. Conversely, if the model predicts stable or tightening spreads, this suggests improving or stable economic conditions. The investor may then increase allocation to corporate bonds to capture higher yields and credit carry, while reducing exposure to safer but lower-yielding assets. In this way, the model enables proactive portfolio adjustment rather than reactive responses after spreads have already moved.

Policy Monitoring and Financial Stability. Policymakers can use the model to see if the business world is going to have financial problems in the future. If predicted credit spreads start to widen before traditional macroeconomic indicators like GDP or unemployment, it could mean that financial conditions are getting tighter and borrowing costs for businesses are going up. Central banks and regulators could use this information to figure out if monetary policy is getting too strict or if they need to provide liquidity support to keep credit markets from getting worse quickly.

Stress Testing and Regulatory Use. In addition, regulators can incorporate the model into stress-testing frameworks by evaluating how projected credit spread movements affect corporate financing costs and balance sheet resilience under adverse economic scenarios. This approach allows policymakers to identify vulnerabilities in the financial system at an early stage and to design more targeted and timely policy responses.

7 Economic Interpretation of the Empirical Results

This part of the chapter will offer an economic explanation regarding the empirical findings. The predicted outcomes from the gradient boosting algorithms will be explained in the context of existing economic models of credit risk and macro-financial relationships. It is important for the outcomes of machine learning algorithms to behave in an expected way in finance.

A main implication of the analysis is the high persistence of the CCC spread series. The high weights assigned by the XGBoost model to the level and long-term transformation terms reflect that a great part of the variance of the spreads series can be attributed to the slowly changing credit cycle components rather than transitory shock components. This finding is in line with the idea that credit risk takes time to build through leverage cycle dynamics and macroeconomic factors, in particular for lower-rated firms.

On the other hand, the LightGBM algorithm is more sensitive to today’s macroeconomic and financial data, and its directional accuracy is higher when adjustment is more rapid. This is a further indication that, while credit spreads have persistence, the location of turning points is more a function of changes in general financial conditions, such as jumps in volatility or reversals in the shape of the yield curve. It would also appear that LightGBM is more adept at picking up regime changes, even if it is less precise when it comes to its level accuracy.

Such differences between the two models illustrate an essential economic trade-off. XGBoost is better at capturing the extent and persistence of spreads, whereas it is more suitable for the task at hand, which includes the areas of valuation, risk analysis, and scenario analysis. LightGBM, on the other hand, is better at capturing the direction of movement and short-run dynamics, which are essential for the monitoring of credit conditions and early warning analyses.

Analysis of debt that is rated CCC highlights the importance of nonlinear techniques. Since these firms are on the edge of default, their credit spread exhibits high degrees of asynchronicity in response to macroeconomic and financial events. Negative events cause more pronounced credit spread widening, while easing causes more gradual credit spread tightening. The success of the GBM approach lends evidence that linear models are inadequate for representing the state-dependent properties of speculative credit risk.

In summary, what emerges is that there is significant value in marrying detailed macro-financial data with flexible machine learning algorithms in terms

8 Future Works and Limitations

Regime Dependence and Structural Breaks. A key limitation of our project is that credit-spread dynamics are not stable over time, and the economic relationships the model learns can change abruptly across market regimes. In normal periods, spreads may respond gradually to macro fundamentals such as unemployment, inflation, and the yield curve. During crisis episodes, spreads can be driven more by liquidity shocks, forced deleveraging, and risk aversion, so variables like volatility and financial conditions dominate. This regime switching creates a structural break problem where a model trained on an “average” of all periods may underperform exactly when forecasts matter

most. It also complicates interpretation, since the same predictor can have different effects depending on whether the market is in a risk-on or risk-off state.

Scope of the Feature Set. Another limitation is the scope of the feature set. While we include U.S. macro and market indicators, credit spreads are influenced by global macro forces such as international growth slowdowns, commodity price shocks, exchange-rate movements, and geopolitical risk, all of which can transmit into U.S. corporate borrowing costs through trade exposure and global capital flows. Because our dataset is largely domestic, the model may miss these external drivers or treat them indirectly through proxies like equity volatility. In addition, several of our “firm-level” inputs are necessarily proxies aggregated at the market level rather than true firm balance sheet measures, which limits our ability to explain cross-sectional credit risk and narrows the interpretation to system-wide credit conditions.

Forecast Horizon Constraints. The forecasting design is also constrained by horizon choice. Our primary target is next-period spread changes, which are practical for short-term risk management; however, they do not fully address longer-term investment decisions that often operate on multi-month or multi-quarter horizons. The drivers of one-month spread moves can differ from the drivers of six-month trends, and models that perform well at one horizon can degrade at others. Extending the framework to multi-horizon forecasting would require changes in labeling, validation strategy, and potentially different model architectures, such as direct multi-output models or horizon-specific models. Similarly, our approach evaluates predictions at a monthly frequency and may not capture faster-moving information that affects spreads at daily or weekly horizons, especially around policy announcements or sudden risk events.

Statistical and Real-Time Forecasting Challenges. Finally, time-series prediction in financial markets has inherent statistical challenges that limit achievable performance. Credit spreads are noisy, exhibit changing volatility over time, and can contain strong persistence, all of which create a risk of overfitting and misleading backtests if the train-test split is not handled carefully. Many features are also released with lags and revisions, meaning that a “real-time” forecasting setup would need to account for publication timing rather than using fully revised historical data. Even with strict time-based splits, forecasts should be interpreted as probabilistic signals rather than precise point estimates. In practice, the model is most useful when combined with uncertainty measures, stress scenarios, and robust risk controls rather than being treated as a standalone decision rule.

9 Overall Assessment

Table 3: XGBoost Metrics

Metric	Train	Validation
MAE	0.0002	0.0003
RMSE	0.000250	0.000859
R^2	1.0000	0.9989
MAPE	0.15%	0.27%
DA	0.6807	0.6449

Table 4: LightGBM Metrics

Metric	Train	Validation
MAE	0.0001	0.0003
RMSE	0.000145	0.000576
R^2	1.0000	0.9995
MAPE	0.08%	0.31%
DA	0.9206	0.7973

Predictive Accuracy and Level Fit. Taken together, the updated graphical diagnostics indicate that both gradient boosting models Figure 5 and Figure 2 deliver exceptionally strong predictive performance for credit spread levels over the validation period. The actual-versus-predicted scatter plots for both XGBoost and LightGBM display an extremely tight alignment along the 45-degree reference line, with no visible systematic bias across the range of observed spreads. This visual evidence is consistent with the near-perfect validation R^2 values reported for both models and confirms that the level of the CCC-10Y spread is captured with high precision.

Dynamic Tracking and Regime Responsiveness. The time-series comparison seen in Figures 6 and 3 plots further reinforce this conclusion. Both models closely track the realized evolution of credit spreads, including prolonged trends and sharp stress episodes. In particular, the LightGBM predictions exhibit very close overlap with the realized series during periods of rapid spread widening and subsequent normalization, suggesting a strong ability to adapt to changing credit regimes. Relative to XGBoost, LightGBM appears slightly more responsive during turning points, which is consistent with its higher directional accuracy on the validation set.

Feature Importance Structure. The feature importance visualizations highlight meaningful structural differences between the two models. XGBoost, displayed in Figure 1, assigns a dominant share of explanatory power to the level and longer-horizon transformations of the credit spread itself, indicating a strong reliance on autoregressive dynamics and persistence in credit conditions. While this contributes to smooth and stable predictions, it may limit sensitivity to broader macro-financial signals.

Macro-Financial Information Utilization. In contrast, the LightGBM, displayed in Figure 4, importance distribution is substantially more dispersed across macroeconomic, market sentiment, and volatility-related features. This broader allocation suggests that LightGBM captures a richer set of conditional interactions between economic fundamentals and credit risk, which likely contributes to its superior directional performance and responsiveness to regime changes.

Overfitting. The choice to accept a greater level of overfitting in the model specification is warranted by the data’s structure and the analysis’s particular goals. The dataset is made up of macroeconomic and financial time series that change slowly and stay the same over time. This is especially true for CCC-rated credit spreads, which only change a little each day and are very path-dependent. In this case, making the model too simple by using shallow trees or heavy regularisation could hide important nonlinear interactions and regime-specific dynamics that are important for how speculative-grade

credit behaves. The models can find small, high-order relationships and turning points in credit conditions that are important for accuracy and early stress detection by letting the tree depth be unlimited, the number of estimators be limited, and the learning rate be low. The modelling goal is not to make broad cross-sectional generalisations, but to make specific predictions for a single credit spread series. This means that finding hidden structure and changes in regime is more important than strict bias-variance optimality. Validation results demonstrating robust out-of-sample fit and elevated directional accuracy further substantiate that the enhanced model flexibility augments practical predictive utility rather than generating extraneous noise, rendering the overfitting trade-off both intentional and economically justifiable.

Implications for Model Choice. Overall, the updated results suggest that while both models are highly effective for level prediction, LightGBM offers a modest but meaningful advantage in capturing dynamic changes in credit spreads and incorporating information from a wider macro-financial feature set. XGBoost remains well suited for applications prioritizing stability and smooth spread estimation, whereas LightGBM appears better aligned with use cases that emphasize early detection of credit regime shifts and changes in market sentiment. These complementary strengths motivate the potential use of ensemble or regime-dependent modeling approaches in future extensions.

10 Conclusion

This study provides a machine learning approach for modeling and forecasting the CCC 10Y credit spread based on a broad spectrum of macro, market, monetary policy, and credit market variables. Through the use of the gradient boosting decision tree approach, this analysis accounts for the non-linear relationships that exist and are important in the context of speculative-grade issuers, where the risk of default and market sentiment are of key significance. XG Boost as well as LightGBM forecasts accurately, establishing the significance of the overall macro-economic environment and the persistence of the spreads on the high-yield credit market. The key takeaway from the empirical findings is that there is an important trade-off between the two modeling methods. XGBoost is very effective at modeling long-run persistence and level, whereas LightGBM is more sensitive to changes in market conditions and has higher directional accuracy, particularly at turning points. From an economics perspective, the results confirm that times of high volatility, lower macroeconomic fundamentals, and tight financial conditions are linked to the widening of credit spreads, whereas improvements in growth and market sentiment are conducive to compression in spreads. The efficacy of these models illustrates that linearity is ineffective at modeling the asymmetric and state-dependent dynamics of CCC-rated credit. However, there exist conditions under which results might be significantly limited for this analysis. The credit spread processes during different market regimes, implying a possible classification error during stressed regimes based on tranquil regime patterns. Additionally, it is primarily a U.S. data set, based on aggregate rather than balance sheet data, implying a constrained inferential domain restricted primarily to forecasting for short investment horizons. Overall, the results demonstrate that combining macro-financial information with flexible machine learning methods provides a powerful and economically intuitive approach to credit spread modeling. While forecasts should be interpreted as probabilistic signals rather than precise predictions, the framework offers valuable insights for risk monitoring, portfolio management, and financial stability analysis, and provides a strong foundation for future extensions incorporating regime dependence, broader data sources, and alternative forecast horizons.

A Appendix

A.1 XGBoost

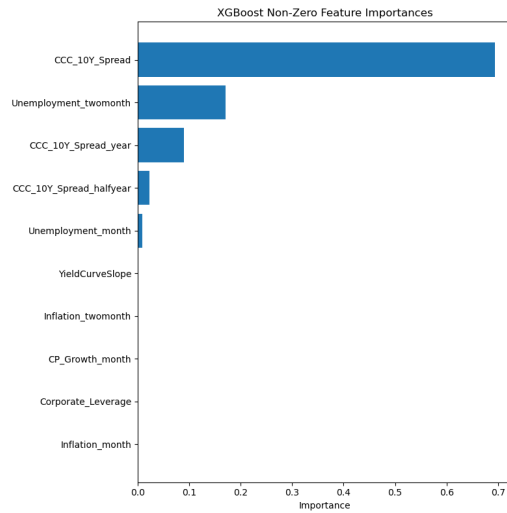


Figure 1: XGBoost Features

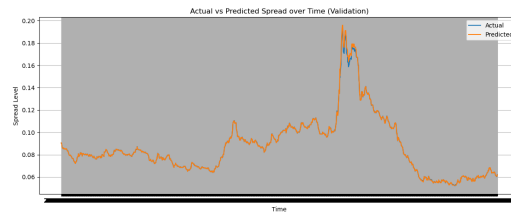


Figure 2: XGBoost Prediction vs Actual Plot

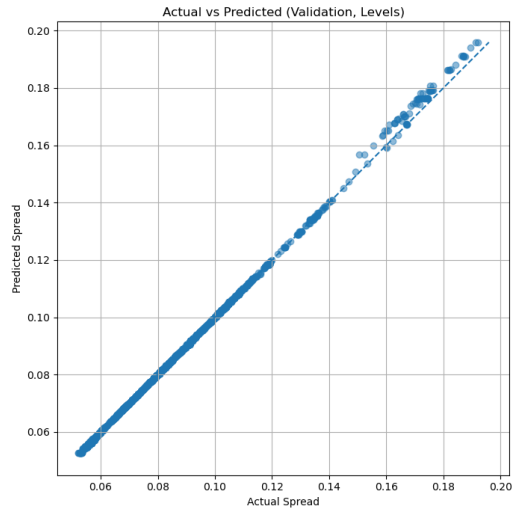


Figure 3: XGBoost Spread accuracy

A.2 LightGBM

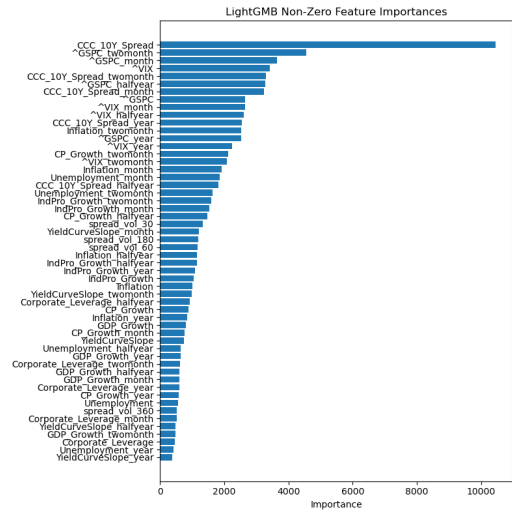


Figure 4: LightGBM Features

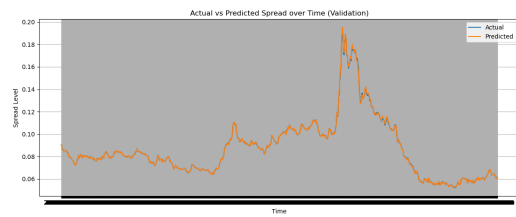


Figure 5: LightGBM Prediction vs Actual Plot

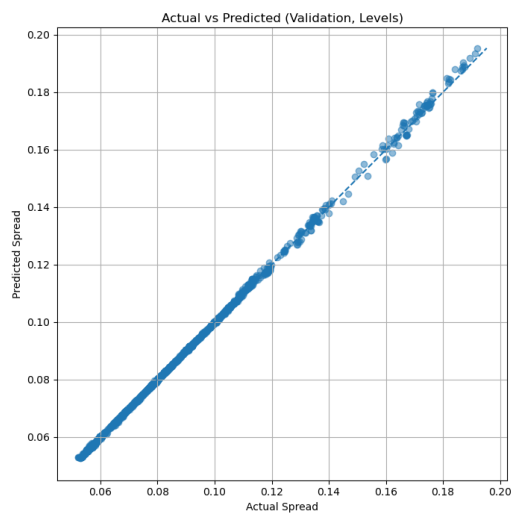


Figure 6: LightGBM Spread accuracy